

SS3 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

FIRST TERM

WEEK 1: REVIEW OF SS1 & SS2 WORK

1.1 INTRODUCTION

Welcome to SS3, the final and most advanced stage of your training. This term, we will move beyond basic installation and troubleshooting to explore emerging technologies, national policies, and the business/environmental context of solar energy.

Before we proceed, we must conduct a thorough review of the foundational knowledge from SS1 and SS2. This recap is essential for understanding the advanced concepts to come.

1.2 CORE CONCEPTS RECAP (SS1 & SS2)

1. CORE COMPONENTS & FUNCTION

- **Solar Panel (PV Module):** The "generator." Composed of photovoltaic cells (usually silicon) that convert sunlight (photons) directly into Direct Current (DC) electricity via the photovoltaic effect.
- **Charge Controller (Solar Regulator):** The "brain" for battery-based systems.
 - **Function:** Regulates the voltage and current coming from the panels to the batteries, preventing overcharging and deep discharge.
 - **Types:**
 - **PWM (Pulse Width Modulation):** Cheaper, less efficient. Acts like a simple switch.
 - **MPPT (Maximum Power Point Tracking):** More expensive, highly efficient (up to 30% more power). It's a smart DC-to-DC converter that matches the panel's voltage to the battery's voltage for optimal charging. *MPPT is the professional standard.*
- **Battery (Energy Storage):** The "storage tank." Stores DC energy from the panels for use at night or during cloudy weather.

- **Key Terms:**
 - **Battery Chemistry:** Deep-Cycle Lead-Acid (Flooded, AGM, Gel), Lithium-ion (LiFePO₄).
 - **Capacity:** Measured in Amp-hours (Ah).
 - **Voltage:** Typically 12V (wired in series for 24V or 48V systems).
 - **DoD (Depth of Discharge):** How much of the battery can be safely used. (Lead-Acid: ~50%; Lithium: 80-100%).
- **Inverter (Power Conversion):** The "engine."
 - **Function:** Converts the low-voltage Direct Current (DC) from the batteries or panels into high-voltage Alternating Current (AC) (e.g., 230V, 50Hz) used by standard home appliances.
 - **Types:** Pure Sine Wave (high quality, for all appliances) vs. Modified Sine Wave (low quality, for simple loads only).

2. SYSTEM TOPOLOGIES (TYPES OF SYSTEMS)

- **Off-Grid (Standalone):**
 - **Components:** Panels + Charge Controller + Batteries + Inverter.
 - **Function:** Completely independent of the national grid (e.g., PHCN/NEPA). Used in rural areas.
 - **Pros:** Total energy independence.
 - **Cons:** Expensive (due to large battery bank); no power if batteries are depleted.
- **On-Grid (Grid-Tied):**
 - **Components:** Panels + Grid-Tied Inverter. (No batteries, no charge controller).
 - **Function:** Works *with* the national grid. Reduces electricity bills.
 - **How:** Prioritizes solar power. If more power is needed, it draws from the grid. If *excess* power is generated, it can be sold *back* to the grid (Net-Metering).
 - **Cons:** *Shuts down during a grid failure* (for safety). Does *not* provide backup power.
- **Hybrid (The Best of Both):**
 - **Components:** Panels + Hybrid Inverter + Batteries (+ Grid/Generator).

- **Function:** The "smart" system. It can charge batteries from both solar and the grid. It can power the home from solar, batteries, or the grid.
- **Modes:**
 1. **Daytime:** Solar powers the home AND charges batteries.
 2. **Evening:** Home runs on battery power.
 3. **Night (Battery Low):** System switches to the grid.
 4. **Grid Failure:** System *instantly* switches to battery/solar power (acts as a UPS).

3. BASIC SYSTEM DESIGN & SIZING

1. **Load Assessment:** The *most critical step*. Calculating the total energy needed per day (measured in Watt-hours, Wh).
2. **Battery Sizing:** Based on Load (Wh) and desired **Days of Autonomy** (how many days of backup power are needed).
3. **Panel Sizing (PV Array):** Based on battery charging needs and the **Peak Sun Hours (PSH)** for the location (e.g., 4.5 - 5.5 PSH in Nigeria).
4. **Inverter Sizing:** Must be larger than the *maximum simultaneous load* (surge load).
5. **Charge Controller Sizing:** Must be able to handle the total current (Amps) from the PV array.

EVALUATION

1. Differentiate between a PWM and an MPPT charge controller. Why is MPPT superior?
2. An on-grid system does not provide backup power. Explain this statement.
3. What is the primary function of a Hybrid Inverter?
4. List the 5 main steps in sizing a new off-grid solar system.

ASSIGNMENT

1. A client has a load of 1,000 Wh per day and wants 2 days of autonomy. If they use a 12V battery system with 50% DoD, what is the *minimum* total Amp-hour (Ah) battery bank required? Show your calculation.

- *(Hint: Total Wh = 1000 * 2. Total Ah = Total Wh / (Voltage * DoD))*



WEEK 2: ADVANCED PV TECHNOLOGIES

2.1 INTRODUCTION

Standard monofacial ("one-sided") solar panels are the norm, but technology is rapidly evolving to increase efficiency and power output from the same or smaller area.

2.2 BIFACIAL PANELS

- **Definition:** A bifacial solar panel is a module that can produce electricity from *both* its front and back sides. The front side captures direct sunlight, while the back side captures reflected and scattered light (known as **albedo**).
- **How They Work:**
 - They use bifacial cells (PERC or HJT cells).
 - The back sheet is transparent (glass or clear backsheet) instead of the standard opaque white plastic.
- **Performance (Bifacial Gain):**
 - They can produce **5% to 30%** more electricity than a monofacial panel of the same wattage rating.
 - This "bifacial gain" depends heavily on **Albedo** (the reflectivity of the surface *behind* the panel).
 - **Albedo Examples:**
 - Dark soil/grass: 10-20% albedo (low gain)
 - Concrete: 15-30% albedo (medium gain)
 - White gravel/paint: 50-80% albedo (high gain)
- **Installation Requirements:**
 - **Mounting:** Must be tilted and raised higher off the ground/roof (at least 0.5m) to allow light to reach the back.
 - **Surface:** The ground or roof should be painted white or covered with reflective material for best results.
- **Pros:** Higher energy yield per panel; better performance in diffuse light (cloudy days).
- **Cons:** Higher initial cost; site-dependent; installation is more complex.

2.3 SOLAR TRACKERS

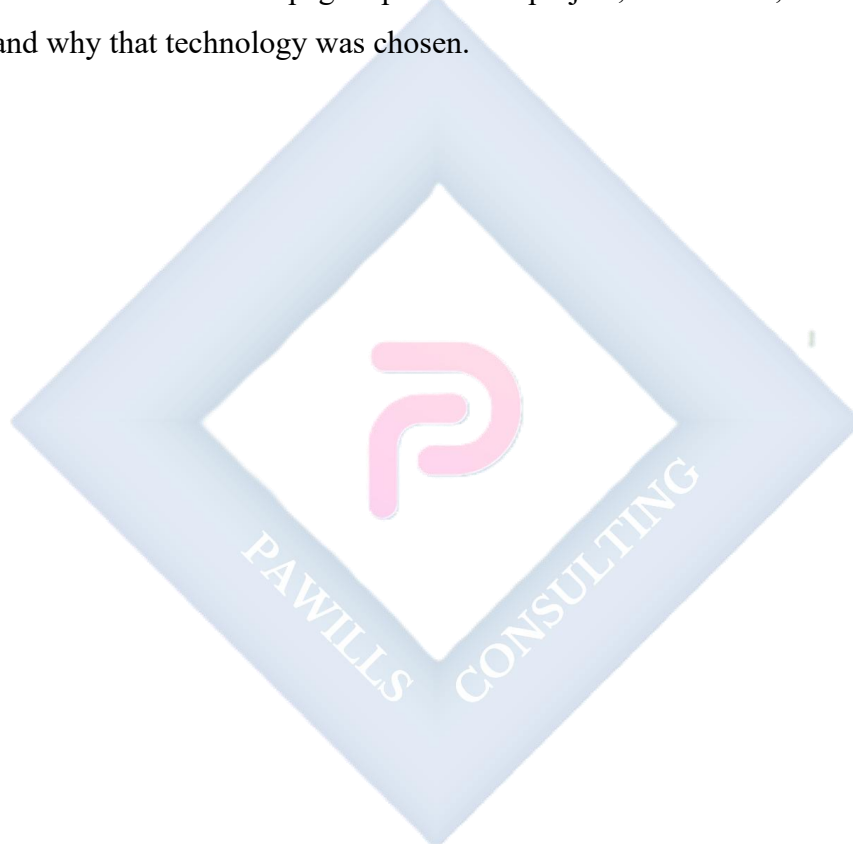
- **Definition:** A solar tracker is a mounting system that automatically orients the solar panels to follow the sun's path across the sky, maximizing sun exposure.
- **Why?** A fixed panel is only at its *perfect* angle to the sun for a few minutes each day. A tracker keeps it at the optimal angle all day.
- **Types of Trackers:**
 1. **Single-Axis Tracker:**
 - Moves the panels on one axis, typically East-to-West, following the sun's daily path.
 - This is the most common and cost-effective type.
 - Can increase energy yield by **25-35%**.
 2. **Dual-Axis Tracker:**
 - Moves the panels on two axes.
 - Tracks the sun's daily E-W path AND the sun's seasonal altitude (its height in the sky, which changes between rainy and dry seasons).
 - Provides the *absolute maximum* power gain (up to **45%** or more) but is complex and expensive.
- **Pros:**
 - Massive increase in energy harvest from the same number of panels.
 - Generates a smoother, longer power curve during the day.
- **Cons:**
 - **High Cost:** Significantly more expensive than fixed mounts.
 - **Complexity & Maintenance:** They have motors, gears, and sensors. These moving parts can fail and require regular maintenance.
- **Application:** Trackers are typically *not* used in residential rooftop solar. They are used in large-scale commercial and utility-scale (solar farm) projects where the extra yield justifies the cost and maintenance.

EVALUATION

1. What is a bifacial panel, and how does it generate more power?
2. Define "Albedo" and explain its importance for bifacial systems.
3. Differentiate between a single-axis and a dual-axis solar tracker.
4. Why are solar trackers not commonly used for residential installations?

ASSIGNMENT

1. Research a commercial project in Nigeria (or West Africa) that uses *either* bifacial panels or solar trackers. Write a one-page report on the project, its location, its capacity (kW or MW), and why that technology was chosen.



WEEK 3: SMART SOLAR SYSTEMS

3.1 INTRODUCTION

A "dumb" solar system works, but you don't know *how* it's working until it fails. A "smart" solar system is connected to the internet, providing real-time data for monitoring, control, and troubleshooting. This is achieved through the **Internet of Things (IoT)**.

3.2 IOT INTEGRATION IN SOLAR

- **What is IoT?** The "Internet of Things" refers to a network of physical devices (like inverters, smart meters, charge controllers) that are embedded with sensors and software, allowing them to "talk" to each other and send data to the internet.
- **How it Works in Solar:**
 1. **Data Collection:** Smart inverters, controllers, or add-on "data loggers" (e.g., a Victron Cerbo GX, SMA Sunny Home Manager) collect performance data (PV voltage, battery SoC, load, grid usage).
 2. **Communication:** This data is sent via Wi-Fi, Ethernet, or a GSM (SIM card) module to a central cloud server run by the manufacturer (e.g., Victron's VRM Portal, Growatt's ShineServer).
 3. **Access:** The user (and the installer, with permission) can then access this data from anywhere in the world using a monitoring app or web portal.

3.3 MONITORING APPS & REMOTE MANAGEMENT

- **Benefits for the End-User:**
 - **Performance Visibility:** See exactly how much energy is being produced and consumed in real-time.
 - **Optimization:** Identify "energy-guzzling" appliances and decide when to use them (e.g., run the water pump when the sun is at its peak).
 - **Fault Alerts:** Get instant notifications (e.g., "Battery Voltage Low," "Grid Failure") on their phone.
- **Benefits for the Installer (Crucial):**

- **Remote Troubleshooting:** A client calls with a problem. The installer can log in, see the system's history and real-time data, and diagnose the fault *without* driving to the site. (e.g., "Sir, the app shows your panels are producing 0 watts. Are they covered in dust, or has someone switched off the breaker?").
- **Preventive Maintenance:** Monitor battery health, panel performance, etc., and schedule maintenance *before* the system fails.
- **System Control:** Remotely change system parameters, start/stop a generator, or update inverter firmware from the office.

3.4 BEYOND MONITORING: SMART ENERGY

- **Load Management:** Smart systems can be programmed to *automatically* turn certain appliances on or off. (e.g., "When battery is 100% and solar is high, turn on the water heater").
- **Smart Grids & VPPs:** In the future, thousands of these smart solar systems can be linked together to form a **Virtual Power Plant (VPP)**, allowing the grid operator to "request" stored battery energy from homes to help stabilize the national grid.

EVALUATION

1. What is the "Internet of Things" (IoT) in the context of a solar installation?
2. List two benefits of a monitoring app for the *end-user*.
3. List two benefits of remote monitoring for the *installer*.
4. Explain how a "smart" system can help an installer troubleshoot a problem remotely.

ASSIGNMENT

1. Download a solar monitoring app (e.g., "VictronConnect" or "ShinePhone" by Growatt) to your phone. Explore the "demo" mode (if available) and write down three specific data points you can see (e.g., "PV Power," "Battery SoC," "Load").

WEEK 4: HYBRID ENERGY SYSTEMS

4.1 DEFINITION

A **Hybrid Energy System (HES)** combines two or more energy generation sources to provide a single, reliable power output. In the SS2 context, a "hybrid inverter" blends Solar + Grid. In this advanced SS3 context, we are referring to blending multiple *local generation* sources, such as:

- Solar PV + Diesel Generator
- Solar PV + Wind Turbine
- Solar PV + Wind + Diesel

The goal is to leverage the strengths of each source and overcome their weaknesses.

4.2 CASE STUDY 1: SOLAR-DIESEL HYBRID (COMMON IN NIGERIA)

- **The Problem:** An off-grid site (like a bank, hospital, or large residence) needs 24/7 power. A solar-only system is too expensive (requires a massive battery bank for autonomy), and a diesel-only system is *extremely* expensive to run (diesel cost, maintenance, noise, fumes).
- **The Solution:** A Solar-Diesel Hybrid.
- **How it Works (The "Brain"):** A smart **Hybrid Inverter** or Power Management System manages all sources.
- **Operational Logic:**
 1. **Priority 1 (Day):** Solar powers the loads directly. Excess solar charges the batteries. The generator is **OFF**.
 2. **Priority 2 (Evening):** Loads run on battery power. The generator is **OFF**.
 3. **Priority 3 (Night/Cloudy):** Battery reaches a low level (e.g., 30% SoC).
 4. **Action:** The inverter automatically sends a "start" signal to the generator. The generator starts, powers the loads, *and* recharges the batteries.
 5. **Shutdown:** Once batteries are charged (e.g., 90%), the inverter shuts the generator **OFF**.
- **Benefits:**

- Massive reduction in diesel consumption (up to 80%).
- Provides 24/7 reliable power.
- Generator runs for fewer hours, extending its lifespan.
- Quiet, clean power for most of the day/night.

4.3 CASE STUDY 2: SOLAR-WIND HYBRID

- **The Concept:** Solar and Wind are highly complementary generation sources.
- **Strengths & Weaknesses:**
 - **Solar:** Works only in the day. Is strongest in the dry season.
 - **Wind:** Can work 24 hours a day. Is often strongest at night and during the rainy/stormy season (when solar is weak).
- **How it Works:**
 - Both the solar panels (via a solar controller) and the wind turbine (via a wind turbine controller) feed DC power to the same battery bank.
 - The hybrid inverter then draws from this battery bank.
- **Challenges:**
 - **Site-Specific:** The location must have *both* good Peak Sun Hours and a good, consistent **average wind speed**. (Most of Nigeria has poor-to-fair wind resources, except for coastal and northern regions).
 - **Complexity:** Wind turbines require more maintenance (moving parts).

EVALUATION

1. What is a Hybrid Energy System (HES)?
2. Describe the 5-step operational logic of a Solar-Diesel hybrid system.
3. What are the major advantages of a Solar-Diesel system over a diesel-only system?
4. Explain why solar and wind are considered "complementary" energy sources.
5. What is the main challenge of installing a Solar-Wind hybrid system in Nigeria?

ASSIGNMENT

1. Imagine you are designing a system for a remote hospital that needs 24/7 power. Why would you recommend a Solar-Diesel hybrid system? Write a one-paragraph justification for the client.



WEEK 5: MIDTERM TEST

(This week is dedicated to a written examination to assess understanding of topics from Weeks 1-4: Advanced PV Tech, Smart Systems, and Hybrid Systems.)



WEEK 6: ENERGY POLICIES IN NIGERIA

6.1 INTRODUCTION

Technology is only one part of the solution. To be a successful installer or solar entrepreneur, you *must* understand the government policies, regulations, and agencies that shape the market.

6.2 KEY AGENCIES

1. Rural Electrification Agency (REA):

- **Mandate:** To provide electricity access to unserved (no grid) and underserved (poor grid) communities, primarily in rural areas.
- **Role:** The REA is one of the *biggest drivers* of the solar market in Nigeria. It implements large-scale projects, often using private solar companies as contractors.
- **Key Programmes:**
 - **Solar Home System (SHS) Deployment:** Provides small "plug-and-play" solar kits to rural homes.
 - **Mini-Grid Development:** Funds the development of small, independent solar-hybrid "mini-grids" to power entire villages or markets.
 - **Energizing Economies/Education:** Solar-powering markets (e.g., Ariaria) and universities.

2. Nigerian Electricity Regulatory Commission (NERC):

- **Mandate:** The primary regulator for the entire Nigerian electricity sector. They set the rules.
- **Role:** NERC creates the regulations that make grid-tied and hybrid solar legally and financially viable.
- **Key Regulations:**
 - **Net-Metering Regulation:** This is the legal framework for on-grid systems. It allows a customer with a grid-tied solar system to "sell" their excess generated electricity back to the local Disco (e.g., Ikeja Electric,

Eko Disco). The Disco must credit the customer's bill. This is the *policy* that makes grid-tied systems financially attractive.

- **Mini-Grid Regulation:** Sets the rules for building and operating mini-grids under 1MW, encouraging private investment.

3. Standards Organisation of Nigeria (SON):

- **Mandate:** To ensure all products sold in Nigeria (including solar components) meet minimum quality standards.
- **Role:** SON-CAP certification is required for imported solar panels, inverters, and batteries to prevent the "dumping" of low-quality, "fake," or unsafe products in the Nigerian market.

EVALUATION

1. What is the full meaning of REA and NERC?
2. Explain the main goal of the Rural Electrification Agency (REA).
3. What is "Net-Metering," and which agency regulates it?
4. Why is the role of the Standards Organisation of Nigeria (SON) important for solar installers?

ASSIGNMENT

1. Visit the official website of the Rural Electrification Agency (REA) of Nigeria. Find one (1) project they have completed. Write down the name of the project, its location, and what it does (e.g., "Solar mini-grid in...")

WEEK 7: MIDTERM BREAK

(No lesson notes for this week.)



WEEK 8: CLIMATE CHANGE & RENEWABLE ENERGY

8.1 THE PROBLEM: CLIMATE CHANGE

- **The Greenhouse Effect:** The Earth is warmed by the sun. Some of this heat is trapped by a "blanket" of gases in the atmosphere (like Carbon Dioxide, CO₂). This is natural and keeps the planet warm.
- **The Cause of Change:** For the last 150 years, burning **fossil fuels** (coal, oil, and natural gas) for electricity, transport, and industry has released *extra* CO₂, making the "blanket" thicker. This is trapping too much heat, causing **Global Warming**.
- **Global Warming vs. Climate Change:** "Global Warming" is the *increase in temperature*. "Climate Change" is the *result* of that increase (e.g., floods, droughts, storms).
- **The Link to Nigeria:**
 - **Diesel Generators:** Millions of diesel generators in Nigeria are a major source of CO₂ emissions, contributing to the problem.
 - **Impact on Nigeria:** Climate change is not a distant problem. It is seen in:
 - **Desertification:** The Sahara desert is expanding south, destroying farmland in the North.
 - **Flooding:** Coastal cities like Lagos and Port Harcourt face severe flooding from rising sea levels.
 - **Erratic Weather:** Unpredictable rainfall patterns that lead to food shortages.

8.2 THE SOLUTION: RENEWABLE ENERGY (SOLAR)

- **Definition:** Renewable energy is energy collected from resources that are naturally replenished, such as sunlight, wind, water (hydro), and geothermal heat.
- **The Role of Solar in Sustainability:**
 - **Zero Emissions:** During operation, a solar panel produces *zero* CO₂. It is a clean source of electricity.

- **Mitigation:** Every solar panel installed *displaces* a dirty source of energy. One solar system for a home can displace a diesel generator, directly *reducing* Nigeria's carbon footprint. This is called **climate change mitigation**.
- **Adaptation:** Solar power can also help communities *adapt* to the effects of climate change.
 - **Example:** As rainfall becomes unreliable (drought), a solar-powered water pump can draw groundwater for irrigation, ensuring food security. This is called **climate change adaptation**.

8.3 SUSTAINABILITY

- **Definition:** Sustainability means meeting the needs of the present without compromising the ability of future generations to meet their own needs.
- **Solar's Role:** Solar energy is the cornerstone of sustainable development because it provides the *energy* needed for economic growth (powering schools, hospitals, businesses) without *destroying* the environment in the process.

EVALUATION

1. Explain the "Greenhouse Effect" in simple terms.
2. What is the difference between "climate change mitigation" and "climate change adaptation"?
3. Give one example of how solar power is used for *mitigation* in Nigeria.
4. Give one example of how solar power is used for *adaptation* in Nigeria.

ASSIGNMENT

1. Write a one-page essay on the topic: "How can solar power help Nigeria achieve its sustainable development goals?" (Focus on health, education, and economic growth).

WEEK 9: ENVIRONMENTAL IMPACTS OF PV SYSTEMS

9.1 THE "CLEAN ENERGY" PARADOX

While solar panels are "zero emission" during their 25-year operational life, they are *not* 100% environmentally friendly. To be a true expert, you must understand the full **Life Cycle Assessment (LCA)** of a PV system.

9.2 LIFE CYCLE STAGES & IMPACTS

1. Manufacturing (Raw Material & Production):

- **Energy:** It takes a large amount of electricity to mine quartz, refine it, and melt it into high-purity silicon wafers. (This is the "Energy Payback Time" - usually 1-3 years).
- **Chemicals:** Strong acids and bases are used to clean and process the silicon.
- **Water:** Significant water is used for cooling and cleaning.

2. Operation (Installation & Use):

- This is the "clean" phase. The main impact is *land use*.
- Utility-scale solar farms can occupy large tracts of land, which can impact local ecosystems (this is less of an issue for rooftop solar).

3. End-of-Life (EoL) - The Big Problem:

- A solar panel has a lifespan of 25-30 years. What happens to it when it "dies"?
- **Solar Waste (E-Waste):** Millions of panels installed today will become millions of tonnes of waste in 25 years.
- **Hazardous Materials:** Panels contain valuable materials (aluminum, glass, silver) but also small amounts of hazardous materials like **lead** (in the solder) and (in older/thin-film panels) **cadmium**.
- **Landfill Risk:** If panels are simply dumped in landfills (like "garbage" or "solar waste"), these toxic materials can **leach** into the soil and groundwater over time, poisoning the water supply.

9.3 THE SOLUTION: SUSTAINABILITY & RECYCLING

- **The "Circular Economy":** The solution is to not treat old panels as "waste" but as a "resource."
- **PV Recycling:**
 - This is a new, specialized industry.
 - **Process:**
 1. The aluminum frame and junction box are removed.
 2. The glass and cell "sandwich" is shredded.
 3. A complex thermal and chemical process is used to separate the glass, silicon, silver, copper, and lead.
 - **Result:** Over 90% of the panel's materials can be recovered and used to make *new* solar panels or other products.
- **The Challenge in Nigeria:**
 - There is currently no large-scale, formal PV recycling facility in Nigeria.
 - This presents a major future environmental risk but also a major **business opportunity** for the first company to establish a recycling plant.

EVALUATION

1. What is a "Life Cycle Assessment" (LCA) for a solar panel?
2. Describe the two main environmental impacts of *manufacturing* a solar panel.
3. What is the main environmental risk of a solar panel at its "End-of-Life"?
4. Why is simply dumping old solar panels in a landfill dangerous?
5. What is the "Circular Economy" in the context of PV waste?

ASSIGNMENT

1. You are a solar installer pitching to an environmentally-conscious client (like a bank). The client says, "I heard solar panels are toxic waste." How would you respond? Write a one-paragraph, honest answer that acknowledges the EoL problem but highlights the recycling solution and the overall net benefit.

WEEK 10: PRACTICAL: SMART SOLAR MONITORING SYSTEM

10.1 OBJECTIVE

To apply the theory from Week 3 by setting up, connecting, and reading data from a smart solar monitoring system.

10.2 MATERIALS

- An installed solar system (PV array, batteries).
- A "smart" inverter or "smart" charge controller (e.g., Victron, Growatt, SMA).
- A data logger / communication device (e.g., Victron Cerbo GX, or an inverter-specific Wi-Fi/GSM dongle).
- A stable internet source (Wi-Fi router or a GSM SIM card).
- A smartphone or laptop with the manufacturer's monitoring app installed (e.g., Victron VRM Portal, ShinePhone).

10.3 PROCEDURE

1. Physical Installation (Safety First!):

- Ensure the main system is operational.
- Power down the inverter and controller.
- Connect the data logger to the inverter/controller using the correct communication cable (e.g., VE.Direct, CAN bus, RS485).
- Connect the data logger to its own power source (if required).
- Power the system back on.

2. Network Configuration:

- Connect the data logger to the internet.
 - **Method A (Wi-Fi):** Access the logger's local hotspot, and configure it to connect to the building's Wi-Fi router.
 - **Method B (GSM):** Insert an activated data SIM card into the logger.

3. Portal Registration:

- On the laptop/phone, go to the manufacturer's monitoring portal website (e.g., vrm.victronenergy.com).
- Create a new "Installer" account.
- Create a new "Installation Site."
- Register the data logger by entering its unique serial number or ID.

4. Data Acquisition:

- Wait 5-10 minutes for the device to "handshake" with the cloud server.
- Open the monitoring app on your phone.
- If successful, you will see a dashboard with live data from the system.

10.4 DATA ANALYSIS (LIVE OBSERVATION)

As a group, observe the dashboard and identify the following key metrics:

- **Solar / PV Harvest:**
 - How much power (Watts) is coming from the panels *right now*?
 - How much energy (kWh) was produced *today*?
- **Battery Status:**
 - What is the Battery Voltage (V)?
 - What is the State of Charge (SoC) as a percentage?
 - Is the battery *charging* (positive Amps) or *discharging* (negative Amps)?
- **Load / Consumption:**
 - How much power (Watts) is the building consuming *right now*?
 - Where is this power coming from? (e.g., 60% from Solar, 40% from Battery).
- **Grid / Generator (if Hybrid):**
 - Is the grid connected?
 - Is the generator running?
 - How much power is being drawn *from* or fed *to* the grid?

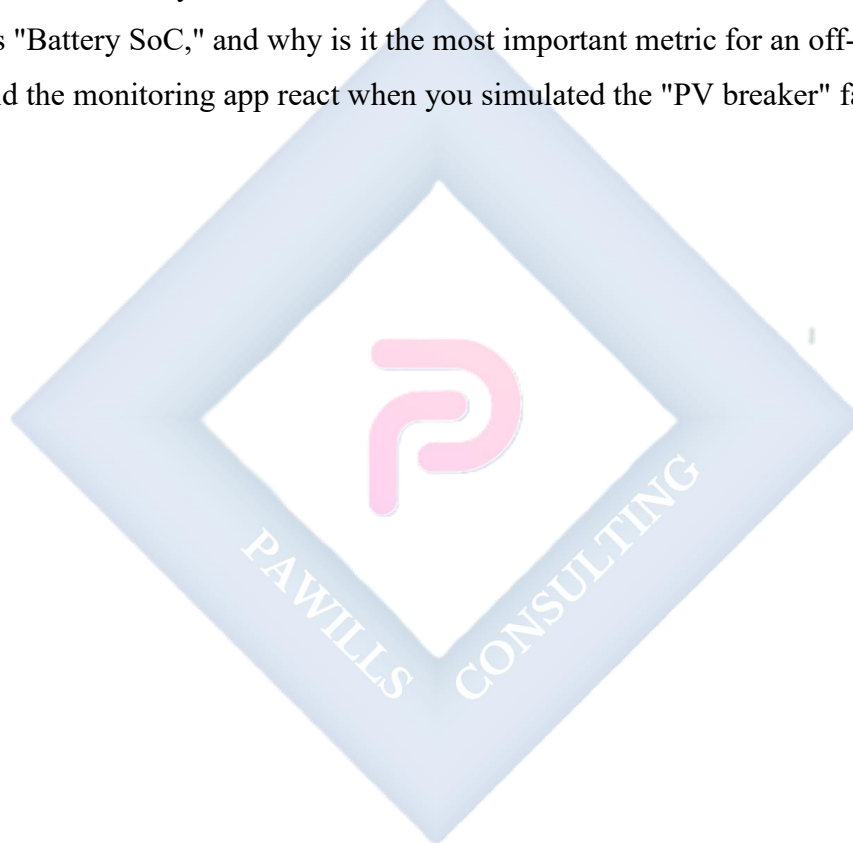
10.5 PRACTICAL APPLICATION (TROUBLESHOOTING)

- **Scenario:** Simulate a fault. Go outside and switch off the "PV-In" breaker.
- **Observation:** Watch the app. What happens?

- The "PV Harvest" will drop to 0W.
- The "Load" will now be supplied 100% by the battery.
- **Conclusion:** This demonstrates how an installer in their office can instantly diagnose a "panels not working" fault, saving a trip to the site.

EVALUATION

1. What is the function of the "data logger" or "Wi-Fi dongle"?
2. List the two main ways a smart inverter connects to the internet.
3. What is "Battery SoC," and why is it the most important metric for an off-grid user?
4. How did the monitoring app react when you simulated the "PV breaker" fault?



WEEKS 11–13: REVISION & EXAMS

- **Revision:** This period is for a comprehensive review of all SS3 First Term topics: Advanced Tech (Bifacial, Trackers), Smart Systems (IoT, Apps), Hybrid Systems (Diesel, Wind), Policy (REA, NERC), and Environment (Climate Change, Recycling).
- **Exam Focus:** Prepare for SSCE-style questions that require *analysis* and *application*.
 - *Example 1:* "Compare and contrast the suitability of a bifacial panel system versus a single-axis tracker system for a large farm in Northern Nigeria."
 - *Example 2:* "A new client in Lagos wants a grid-tied system to save money, but also wants backup power. What system (Off-grid, On-grid, or Hybrid) would you recommend, and why?"
 - *Example 3:* "Explain the role of the REA and NERC in promoting the growth of the Nigerian solar market."
- **Final Examinations:** First Term Examination for SS3.



SS3 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

SECOND TERM

WEEK 1: PREPARATION FOR SSCE – REVIEW OF KEY TOPICS

1.1 INTRODUCTION: THE FINAL PUSH

Welcome to your final practical term. The goal for the next 10 weeks is singular: to ensure you are 100% prepared to excel in your SSCE (WAEC/NECO) Theory, Objective, and Practical examinations. This term bridges the gap between knowing *about* solar and *being* a solar technician.

We will achieve this through intensive review, analysis of past questions, and a capstone "Mock Installation Project" that will test all your skills.

1.2 CONSOLIDATED SS1-SS3 KNOWLEDGE MAP

This is your master checklist for revision. Ensure you can confidently explain every item.

A. SS1 FOUNDATIONS (THE "WHAT")

- **Safety & Tools:**
 - **PPE:** Full meaning and use (Helmet, Gloves, Goggles, Boots).
 - **Tools:** Identification and specific use of:
 - **Electrical:** Multimeter, Clamp Meter, Insulation Tester.
 - **Hand:** Pliers, Crimping Tool (MC4), Spanners, Wire Stripper.
 - **Power:** Drill, Angle Grinder.
- **Core Electrical Principles:**
 - Ohm's Law ($V=IR$), Power Law ($P=IV$).
 - Difference between Voltage (V), Current (I), Resistance (R), and Power (W).
 - Difference between Energy (Wh/kWh) and Power (W/kW).
 - AC vs. DC.
 - Series vs. Parallel circuits (know the effect on V and I for panels and batteries).

- **Basic Components:**

- **PV Panels:** Monocrystalline vs. Polycrystalline. Key specs (V_{oc} , I_{sc} , V_{mp} , I_{mp}).
- **Batteries:** Deep-Cycle vs. Shallow-Cycle. Lead-Acid (Flooded, AGM, Gel) vs. Lithium-ion ($LiFePO_4$).
- **Inverters:** Function (DC to AC). Pure Sine Wave vs. Modified Sine Wave.
- **Charge Controllers:** Function (protects battery). PWM vs. MPPT.

B. SS2 SYSTEM DESIGN & INSTALLATION (THE "HOW")

- **System Sizing (The 5 Steps):**

1. **Load Assessment:** Calculating total daily Watt-hours (Wh).
2. **Inverter Sizing:** Based on total *Power* (W) + surge.
3. **Battery Bank Sizing:** Based on *Energy* (Wh), Voltage, Days of Autonomy, and Depth of Discharge (DoD).
4. **PV Array Sizing:** Based on Battery Bank size, System Voltage, and Peak Sun Hours (PSH).
5. **Charge Controller Sizing:** Based on PV Array current (I_{sc}) and voltage (V_{oc}).

- **System Topologies (Types):**

- **Off-Grid:** Components, pros, cons. (Relies on batteries).
- **On-Grid (Grid-Tied):** Components, pros, cons. (No batteries, requires grid, Net-Metering).
- **Hybrid:** Components, pros, cons. (Combines all sources: Solar, Battery, Grid/Generator).

- **Installation Process:**

- Site Assessment, Mounting (Roof vs. Ground), Wiring, Commissioning.

- **Maintenance & Troubleshooting:**

- **Maintenance:** Panel cleaning, terminal tightening, battery specific gravity/voltage checks.
- **Troubleshooting:** Logical fault finding (Inverter? Battery? Controller? Panels?).

C. SS3 ADVANCED CONCEPTS (THE "WHY")

- **Advanced Tech:** Bifacial Panels (Albedo), Solar Trackers (Single/Dual-Axis).
- **Smart Systems:** IoT, Monitoring Apps, Remote Troubleshooting.
- **Hybrid Systems:** Solar-Diesel logic (generator auto-start).
- **Policy:** REA (Rural Electrification), NERC (Regulation, Net-Metering), SON (Standards).
- **Environment:** Climate Change (Mitigation/Adaptation), PV Life Cycle (E-Waste, Recycling).

EVALUATION

1. Explain the difference between Energy (kWh) and Power (kW).
2. What is the primary operational difference between an On-Grid and an Off-Grid system?
3. Why is a Pure Sine Wave inverter essential for sensitive electronics?

ASSIGNMENT

1. Go through the entire "Knowledge Map" above.
2. Identify and list your THREE (3) weakest areas.
3. Spend the rest of this week conducting intensive personal revision on those three topics.

WEEK 2: SSCE PAST QUESTIONS – THEORY (WORKED SOLUTIONS)

2.1 INTRODUCTION

The theory exam tests your *understanding* of the concepts. Questions are designed to see if you can define, differentiate, explain, and calculate. The key verbs are **List, State, Differentiate, Explain, Describe, Calculate.**

2.2 WORKED EXAMPLES (SSCE-STYLE THEORY QUESTIONS)

Question 1: (DEFINITIONS) (a) What is a Photovoltaic (PV) system? (b) List the four (4) core components of a standalone (off-grid) solar system.

- **Model Answer:** (a) A Photovoltaic (PV) system, or solar power system, is an assembly of components designed to convert sunlight (photons) directly into electricity (DC voltage) using the photovoltaic effect, which is then stored or used to power electrical loads. (b) The four core components of a standalone system are: 1. Solar Panels (PV Modules) 2. Charge Controller 3. Battery (or Battery Bank) 4. Inverter

Question 2: (DIFFERENTIATION) Differentiate between a "Monocrystalline" and a "Polycrystalline" solar panel.

- **Model Answer:** *A table is the best way to answer this.* | Feature | Monocrystalline Panel | Polycrystalline Panel | | :--- | :--- | :--- | | **Cell Appearance** | Uniform black colour, rounded edges. | Blue, "marbled" appearance, perfectly square cells. | | **Manufacturing** | Made from a single, pure silicon crystal. | Made from multiple silicon fragments melted together. | | **Efficiency** | Higher efficiency (18-23%). | Lower efficiency (15-18%). | | **Space** | Produces more power in a smaller area. | Requires more space for the same power output. | | **Cost** | More expensive to produce. | Less expensive to produce. |

Question 3: (EXPLANATION) Explain the function of a charge controller in a solar installation and state the two main types.

- **Model Answer:**

- **Function:** The primary function of a charge controller (or solar regulator) is to protect the battery bank. It is the "brain" of the battery system. It regulates the voltage and current flowing from the solar panels to the batteries to prevent **overcharging** (which can boil the battery) and to prevent **deep discharge** (by cutting off the load when the battery is too low).
- **Types:** The two main types are:
 1. **PWM (Pulse Width Modulation)**
 2. **MPPT (Maximum Power Point Tracking)**

Question 4: (CALCULATION) A client's load assessment shows a total daily energy requirement of 2,100 Watt-hours (Wh). The client wants a 12V battery system with 1 day of autonomy and a 50% Depth of Discharge (DoD). Calculate the minimum required capacity (in Amp-hours) of the battery bank.

• **Model Answer:**

- **Given:**
 - Daily Energy (Load) = 2,100 Wh
 - Days of Autonomy = 1 day
 - System Voltage = 12 V
 - Depth of Discharge (DoD) = 50% or 0.5
- **Formula:**
 - $\text{Total Battery Capacity (Ah)} = (\text{Daily Energy} \times \text{Days of Autonomy}) / (\text{System Voltage} \times \text{DoD})$
- **Calculation:**
 - $\text{Total Wh needed} = 2,100 \text{ Wh} \times 1 = 2,100 \text{ Wh}$
 - $\text{Ah Capacity} = 2,100 \text{ Wh} / (12 \text{ V} \times 0.5)$
 - $\text{Ah Capacity} = 2,100 / 6$
 - $\text{Ah Capacity} = 350 \text{ Ah}$
- **Answer:** The client needs a minimum battery bank of **350 Ah at 12V**. (Practical: This could be two 200Ah 12V batteries in parallel).

EVALUATION

1. Explain the difference between "Peak Power" (W) and "Daily Energy" (Wh).
2. What is the primary risk of *not* installing a charge controller?

ASSIGNMENT

1. Find an SSCE past question paper for this subject.
2. Select and answer FIVE (5) theory questions.
3. Practice writing your answers clearly and showing all calculations.



WEEK 3: SSCE PAST QUESTIONS – PRACTICAL CASE STUDIES

3.1 INTRODUCTION

The "Practical" exam is often a written paper that presents you with a **case study** or a **fault scenario**. It tests your ability to *apply* your knowledge to solve a real-world problem. It tests your logical thinking, safety consciousness, and design skills.

3.2 WORKED EXAMPLES (SSCE-STYLE CASE STUDIES)

Case Study 1: (Sizing & Justification)

- **Question:** A small rural school needs to power a computer lab with the following loads:
 - 4 Computers (150W each)
 - 8 LED Bulbs (10W each)
 - 1 Printer (80W, used intermittently)
 - All loads must run for 5 hours per day.
 - The school has no grid power.
- **Tasks:**
 - (a) Calculate the total daily energy load (Wh).
 - (b) What *type* of inverter MUST be used, and why?
 - (c) Calculate the *minimum* power rating (W) for the inverter.
- **Model Answer:**
 - **(a) Total Daily Energy Load:**
 - Computers: $4 \times 150\text{W} \times 5\text{h} = 3,000 \text{ Wh}$
 - Bulbs: $8 \times 10\text{W} \times 5\text{h} = 400 \text{ Wh}$
 - Printer: $1 \times 80\text{W} \times 5\text{h} = 400 \text{ Wh}$ (*Note: This is a safe, high estimate*)
 - **Total Load = $3,000 + 400 + 400 = 3,800 \text{ Wh/day}$**
 - **(b) Inverter Type & Justification:**
 - **Type:** A **Pure Sine Wave (PSW)** inverter must be used.
 - **Justification:** Computers, printers, and modern electronics are *sensitive* loads. A Modified Sine Wave inverter produces a "blocky" AC signal that can damage or destroy their power supplies. A Pure Sine Wave inverter

produces a clean, smooth AC signal identical to the grid, which is safe for all appliances.

○ **(c) Minimum Inverter Power Rating (W):**

- *First, calculate the total simultaneous power:*
- Computers: $4 \times 150\text{W} = 600\text{W}$
- Bulbs: $8 \times 10\text{W} = 80\text{W}$
- Printer: $1 \times 80\text{W} = 80\text{W}$
- **Total Power = $600 + 80 + 80 = 760\text{W}$**
- *Then, add a safety/surge factor (e.g., 25%):*
- Minimum Rating = $760\text{W} \times 1.25 = 950\text{W}$
- **Answer:** A standard **1000W** (or 1kW) inverter would be the minimum suitable choice.

Case Study 2: (Troubleshooting)

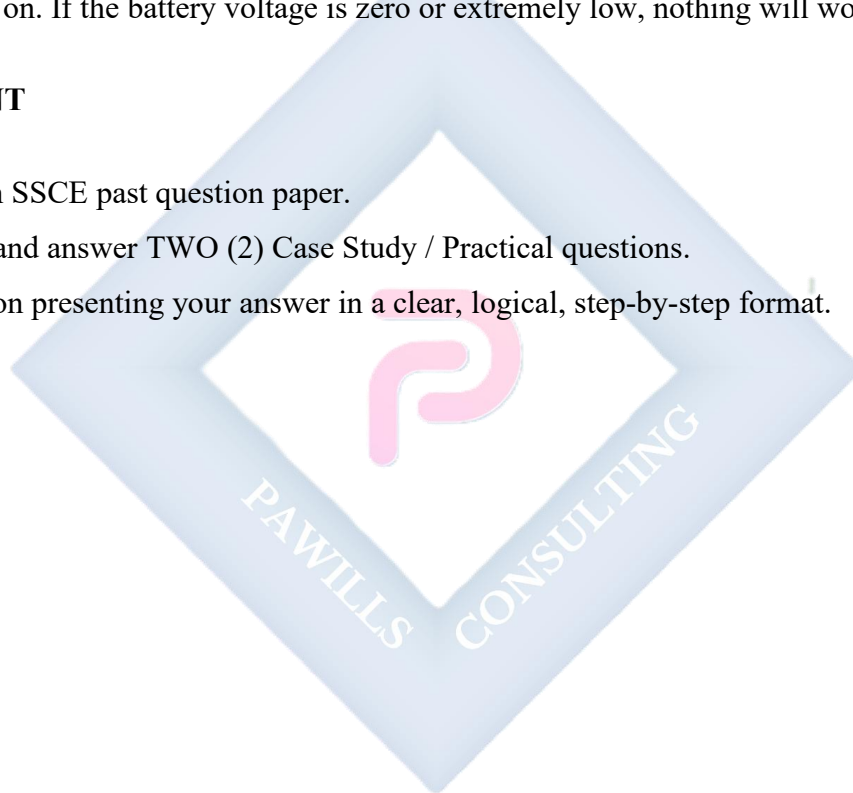
- **Question:** You receive a call from a client. Their solar system (installed 2 years ago) suddenly stopped working at 8:00 PM. The inverter is beeping with a "Low Battery" alarm, and the lights are off.
- **Task:** List, in logical order, four (4) *possible causes* of this fault.
- **Model Answer:**
 - **Possible Causes:**
 1. **Unusual High Load:** The client may have used more energy than usual during the day (e.t., had visitors, used a high-power appliance like an iron), draining the batteries faster than normal.
 2. **Poor Charging (Recent):** There may have been several days of heavy rain or clouds, preventing the solar panels from fully charging the batteries.
 3. **PV System Fault:** A part of the charging system may have failed (e.g., a PV breaker tripped, a panel string is faulty, the charge controller is damaged). This means the battery did not charge at all during the day.
 4. **Battery Degradation:** The batteries are 2 years old. They may be failing and losing their capacity (they can no longer hold a full charge).

EVALUATION

1. What is the functional difference between an inverter's Power rating (W) and its Energy capacity (Wh)? (This is a trick question).
 - *Answer:* An inverter only has a Power (W) rating. *Batteries* have an Energy (Wh) capacity.
2. A client's inverter screen is blank and the lights are off. What is the *very first* component you should check with your multimeter?
 - *Answer:* The **battery bank voltage**. The inverter needs the battery to power itself on. If the battery voltage is zero or extremely low, nothing will work.

ASSIGNMENT

1. Find an SSCE past question paper.
2. Select and answer TWO (2) Case Study / Practical questions.
3. Focus on presenting your answer in a clear, logical, step-by-step format.



WEEK 4: MOCK INSTALLATION PROJECT I – DESIGN

4.1 THE PROJECT BRIEF (CLIENT REQUEST)

- **Client:** Mr. Ade, for his new 2-bedroom bungalow in a rural area (no grid).
- **Goal:** To design a complete off-grid system.
- **Load Assessment (provided by client):**
 - 1 x Refrigerator (120W, 12h/day compressor cycle)
 - 10 x LED Bulbs (9W each, 6h/day)
 - 2 x Fans (60W each, 8h/day)
 - 1 x TV (100W, 5h/day)
 - 1 x Water Pump (0.5hp, approx. 375W, 1h/day)
- **Requirements:**
 - System Voltage: **24V** (Specified by instructor for this project).
 - Autonomy: **1.5 Days**.
 - Battery DoD: **50%**.
 - Peak Sun Hours (PSH): **5 hours**.
- **Your Task (in Groups):** Design the full system on paper. You must calculate and specify the exact size/rating for all 5 major components.

4.2 PHASE 1: LOAD ASSESSMENT & INVERTER SIZING

- **Step 1: Calculate Total Daily Energy (Wh)**
 - Fridge: $120\text{W} \times 12\text{h} = 1,440 \text{ Wh}$
 - Bulbs: $10 \times 9\text{W} \times 6\text{h} = 540 \text{ Wh}$
 - Fans: $2 \times 60\text{W} \times 8\text{h} = 960 \text{ Wh}$
 - TV: $1 \times 100\text{W} \times 5\text{h} = 500 \text{ Wh}$
 - Pump: $1 \times 375\text{W} \times 1\text{h} = 375 \text{ Wh}$
 - **TOTAL DAILY ENERGY = 3,815 Wh/day**
- **Step 2: Calculate Total Power (W) & Size Inverter**
 - Total *Running* Load: $120\text{W} + (10 \times 9\text{W}) + (2 \times 60\text{W}) + 100\text{W} + 375\text{W} = 805\text{W}$
 - Surge: The water pump (375W) will have a surge. (Surge $\approx 3 \times 375\text{W} = 1,125\text{W}$).

- **Inverter Sizing:** The inverter must handle the *surge + other loads*. A 2,000W (2kW) or 2.5kVA inverter is a safe choice.
- **Specification:** 1 x 2.5kVA / 2000W, 24V Pure Sine Wave Inverter

4.3 PHASE 1 (CONT.): BATTERY, PV ARRAY, & CONTROLLER SIZING

- **Step 3: Size Battery Bank (Ah)**

- Formula: $Ah = (Total\ Energy \times Autonomy) / (Voltage \times DoD)$
- $Ah = (3,815\ Wh \times 1.5\ days) / (24\ V \times 0.5)$
- $Ah = 5,722.5 / 12$
- $Ah = 476.8\ Ah$
- **Specification:** 477 Ah @ 24V. (Practical: 4 x 250Ah 12V batteries, wired in 2S2P configuration).

- **Step 4: Size PV Array (W)**

- *First, find target charge (Wh):* 3,815 Wh
- *Account for PSH:* $3,815\ Wh / 5\ PSH = 763\ W$
- *Account for system losses (e.g., 25%):* $763\ W \times 1.25 = 954\ W$
- **Specification:** ~1000W (1kW) PV Array. (Practical: 3 x 350W panels = 1050W).

- **Step 5: Size Charge Controller (A)**

- *First, find max current from panels (Isc):* (Using 3x 350W panels. Assume $I_{sc} \approx 10A$ each).
- *If wired in parallel (not good for 24V):* $10A + 10A + 10A = 30A$.
- *If wired in series:* Voltage is too high.
- *Best way:* $Controller\ Amps = Total\ Panel\ Power / System\ Voltage$
- $Amps = 1050W / 24V = 43.75\ A$
- *Add safety factor (25%):* $43.75\ A \times 1.25 = 54.7\ A$
- **Specification:** 1 x 60A MPPT Charge Controller (min. 100V Voc rating)

4.4 FINAL BILL OF MATERIALS (BOM) - DESIGN PHASE

- **Inverter:** 1 x 2.5kVA, 24V, Pure Sine Wave
- **Batteries:** 4 x 250Ah 12V Deep-Cycle (AGM or Gel)

- **Panels:** 3 x 350W Monocrystalline
- **Controller:** 1 x 60A MPPT
- **Misc:** MC4 connectors, 16mm² battery cables, 6mm² panel cables, breakers (DC/AC), surge protector, mounting rails.

EVALUATION

1. Why was a 24V system chosen over a 12V system for this load?
 - *Answer:* The load is large. At 24V, the current is *half* what it would be at 12V. This allows for thinner, cheaper cables and reduces resistive losses.
2. What does "2S2P" mean for wiring the 4 batteries?
 - *Answer:* It means "Two in Series, Two in Parallel." You make two sets of 24V (by wiring two 12V in series), and then connect those two sets in parallel to double the capacity (Ah).

ASSIGNMENT

1. In your groups, create a full Bill of Quantities (BOQ) for this project. Research the approximate current market price in Naira for each of the 5 main components and calculate a total material cost.

WEEK 5: MIDTERM TEST

(This week is dedicated to a written midterm test, covering all SSCE revision topics from Weeks 1-4, with a strong focus on calculations and case study analysis.)



WEEK 6: MOCK INSTALLATION PROJECT II – INSTALLATION & COMMISSIONING

6.1 OBJECTIVE

To physically and safely install the system designed in Week 4, following all professional standards and safety protocols.

6.2 PHASE 2: INSTALLATION PROCEDURE (PRACTICAL)

- **A. SAFETY FIRST (PPE):** All students must wear boots, gloves, and (if on a roof) helmets.
- **B. MOUNTING & PANEL INSTALLATION:**
 1. Assemble ground mounts or fix roof rails (using L-feet and sealant).
 2. Haul panels to the roof (if applicable).
 3. Secure panels to the rails using mid-clamps and end-clamps.
 4. **Grounding:** Run a grounding wire connecting the frame of *every single panel* and the rails to a dedicated earth rod.
- **C. DC WIRING & "THE BOX":**
 1. **Panel Wiring:** Connect the 3x 350W panels. (As they are 60-cell panels, $V_{mp} \approx 38V$. Wiring all 3 in series gives $V_{mp} \approx 114V$. This is too high for a 100V controller. Wiring them in parallel gives $I_{sc} \approx 30A$, which is fine for the 60A controller).
 2. *Decision:* Wire all 3 panels in **Parallel** using MC4 branch connectors.
 3. Run the main (+ and -) 6mm² solar cable from the roof to the "DC Combiner Box."
 4. **Inside the Box:** Install the DC breakers (one for the PV array) and the DC Surge Protection Device (SPD).
- **D. BATTERY BANK & INVERTER ROOM:**
 1. Install the inverter and charge controller on the wall (in a cool, ventilated space).
 2. Build the battery bank (4 x 12V batteries) in the 2S2P configuration as designed.
 3. Crimp lugs onto the thick (16mm² or 25mm²) battery cables.

4. Install the main DC isolator (breaker) between the battery bank and the inverter/controller.

6.3 PHASE 2 (CONT.): COMMISSIONING (ORDER OF OPERATIONS)

This is the most critical safety procedure. Doing this in the wrong order can **destroy** the equipment.

ALL BREAKERS MUST BE IN THE "OFF" POSITION TO START.

1. **Step 1 (Controller to Battery):** Connect the Battery-Out terminals of the Charge Controller to the Battery busbar/isolator.
2. **Step 2 (Power on Controller):** Turn ON the Battery-to-Controller breaker.
 - *Result:* The Charge Controller boots up, reads the 24V system, and its screen turns on. It now "knows" it is in a 24V system.
3. **Step 3 (Inverter to Battery):** Connect the main Inverter (DC Input) cables to the Battery busbar/isolator.
4. **Step 4 (PV to Controller):** Connect the PV-In cables (from the roof) to the PV-In terminals on the Charge Controller.
5. **Step 5 (Power on PV):** Turn ON the PV-In breaker.
 - *Result:* The controller will "see" the voltage from the panels and (if sunny) will begin charging the batteries.
6. **Step 6 (Inverter to Load):** Connect the AC-Out of the inverter to the building's distribution board (load).
7. **Step 7 (Final Power On):** Turn ON the Inverter-to-Battery breaker, then turn ON the Inverter itself.
 - *Result:* The system is now live. The inverter will start producing 230V AC power for the loads.

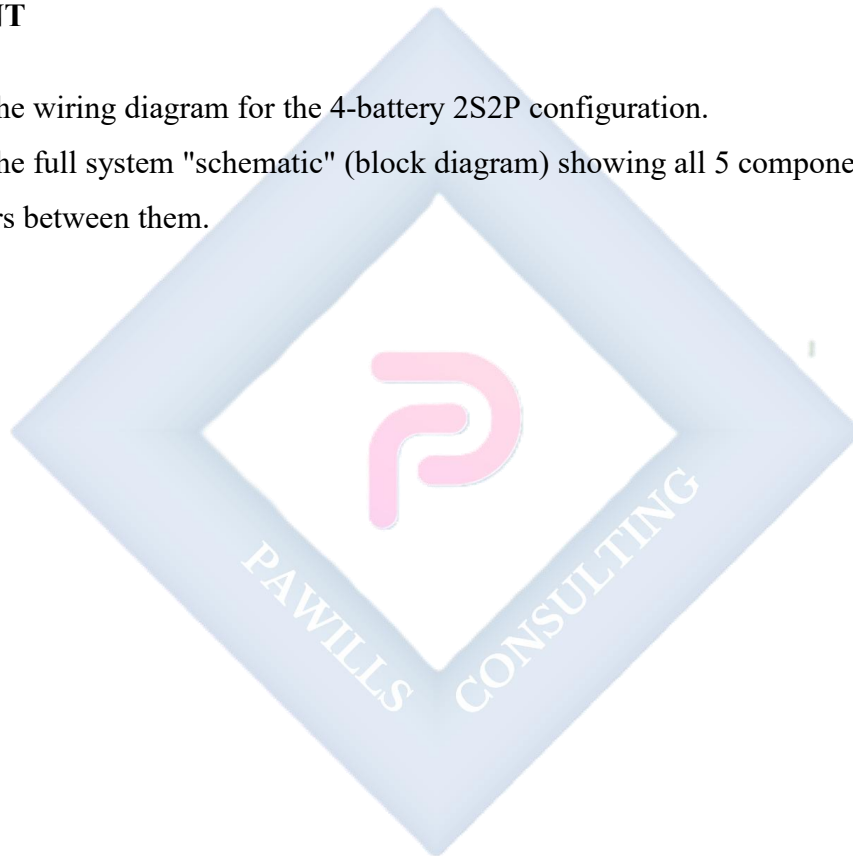
EVALUATION

1. Why *must* you connect the charge controller to the battery *before* connecting it to the panels?

- *Answer:* The controller needs to read the battery voltage (12V? 24V? 48V?) *first* so it knows what voltage to output. If you connect the panels first, the high voltage (e.g., 100V) can confuse or damage the controller.
2. What is the purpose of "grounding" the panel frames?
- *Answer:* For safety, to prevent electric shock. If a wire inside the panel accidentally touches the frame, the current will be safely directed to the earth, tripping a breaker, instead of electrocuting someone who touches it.

ASSIGNMENT

1. Draw the wiring diagram for the 4-battery 2S2P configuration.
2. Draw the full system "schematic" (block diagram) showing all 5 components and the breakers between them.



WEEK 7: MIDTERM BREAK

(No lesson notes for this week.)



WEEK 8: MOCK INSTALLATION PROJECT III – TROUBLESHOOTING & MAINTENANCE

8.1 OBJECTIVE

To test the logical troubleshooting skills of the students by simulating common faults on the newly installed project.

8.2 PHASE 3: FAULT SIMULATION & DIAGNOSIS (PRACTICAL)

The instructor will secretly create a fault. Students must use their tools and senses to find it.

- **FAULT 1: "System is dead. Inverter is off. Controller is off."**
 - **Diagnosis:** This means there is NO power from the batteries.
 - **Tool:** Multimeter (DC Voltage mode).
 - **Steps:**
 1. Check main battery isolator (is it tripped?).
 2. Check voltage at the inverter's DC input (will be 0V).
 3. Check voltage at the main battery terminals.
 4. *Finding:* A student has disconnected the main negative terminal from the battery bank.
 - **Solution:** Reconnect the terminal.
- **FAULT 2: "System works, but battery is draining. No charge from solar."**
 - **Diagnosis:** The link between the panels and the controller is broken.
 - **Tool:** Multimeter (DC Voltage & Amps mode).
 - **Steps:**
 1. Observe controller (shows 0W from PV).
 2. Check PV input breaker (is it tripped?).
 3. *Finding:* The instructor has tripped the PV-In breaker in the DC box.
 4. (If breaker wasn't the fault): Go to the roof, check Voc of the array. Check Isc. The fault could be a bad MC4 connection.
 - **Solution:** Reset the breaker.

- **FAULT 3: "Inverter beeps with 'Overload' alarm when client turns on the water pump."**
 - **Diagnosis:** The *surge* load of the pump is higher than the inverter's capacity.
 - **Cause:** The inverter was sized *incorrectly* for the surge (Week 4 design flaw) OR the client has bought a *different, larger* pump.
 - **Solution:**
 1. **Immediate:** Tell the client they cannot run the pump.
 2. **Long-Term:** The inverter must be upgraded to a larger model (e.g., 3.5kW) OR the pump must be replaced with a "soft-start" or lower-power model.
- **FAULT 4: "Client complains of low battery every night, even on sunny days."**
 - **Diagnosis:** The load is higher than the design, OR the PV array is underperforming.
 - **Tool:** Clamp Meter (DC Amps).
 - **Steps:**
 1. Check PV array for shading (a new tree? a neighbour's new wall?).
 2. Check panels for extreme dust/dirt.
 3. Check array current (Amps) at the controller. If it's 50A at noon, panels are good.
 4. *Finding:* Put a clamp meter on the AC-Out. Discover the client's "120W" fridge is actually an old, faulty 300W model.
 - **Solution:** The "Load Assessment" in Week 4 was wrong. The client must replace the faulty fridge or the PV array must be expanded.

EVALUATION

1. What tool is your *first* choice for any "dead system" fault? (Multimeter).
2. What is the difference between an "Overload" alarm and a "Low Battery" alarm?
 - *Answer:* "Overload" is a *Power* (W) problem (too much at once). "Low Battery" is an *Energy* (Wh) problem (all power is used up).
3. What is the most likely cause of reduced panel performance (low Amps) on a clear day?
 - *Answer:* Dust/dirt on the panels, or partial shading.

ASSIGNMENT

1. Write a simple one-page "Handover Document" for the client (Mr. Ade). It should explain (in simple terms) what the system can and cannot power, and list 3 basic maintenance checks the client should do (e.g., check for shade, clean panels, check for error lights).



WEEK 9: MOCK SSCE

(This week is dedicated to a full-scale Mock Examination (Theory, Objective, and Practical) under strict exam conditions. This will be set and graded by the teacher to simulate the real SSCE.)



WEEK 10: EXAM PREP & COACHING

10.1 ANALYSIS OF MOCK EXAM

- This session involves the teacher returning the mock exam scripts.
- **Group Analysis:** Go through the most common mistakes.
 - *Did students mix up Power (W) and Energy (Wh) in calculations?*
 - *Did students forget the safety factor in inverter sizing?*
 - *Were troubleshooting steps logical?*
 - *Were definitions clear and complete?*
- **Individual Coaching:** Students receive one-on-one feedback on their specific weak areas.

10.2 MASTERING EXAM STRATEGY

- **Objective Questions:**
 - **Time:** Do *not* spend more than 1 minute per question.
 - **Process:** Read the question, then the options. Eliminate the 2 *impossible* answers first. Choose the *best* answer from the remaining 2.
 - **Guessing:** If you are unsure, *always* make an educated guess. Never leave an objective question blank.
- **Theory / Essay Questions:**
 - **Verbs:** Understand the question. "List" = bullet points. "Explain" = paragraphs.
 - **Calculations:** Show *all* your work (Formula, Substitution, Answer). You get marks for the *method* even if the final number is wrong.
 - **Clarity:** Write neatly. Use a table for "Differentiate" questions. Use a numbered list for "Describe procedure" questions.
- **Practical / Case Study:**
 - **Safety First:** Always mention safety (PPE, breakers) in any installation or troubleshooting question. It's an easy mark.
 - **Logic:** Think like a technician. What is the *simplest* and *most likely* fault? Check that first. (e.g., Don't climb to the roof before checking the breaker).

- **Be Specific:** Don't just say "check the battery." Say "use a multimeter to check the battery's DC voltage at the terminals."

EVALUATION

1. What was your single biggest mistake in the Mock Exam?
2. How do you get partial marks on a calculation question even if your math is wrong?

ASSIGNMENT

1. Based on your Mock Exam results, create a final, one-page "Personal Study Guide" focusing *only* on the 3-5 topics you still find difficult. This is your final study sheet.

WEEKS 11–13: REVISION & EXAMS

- **Final Review:** This period is for final, independent student revision of all SS1-SS3 materials, using the Knowledge Map (Week 1), Past Questions (Weeks 2-3), and Mock Exam feedback (Week 10) as guides.
- **Good luck in your SSCE!**

SS3 SOLAR PHOTOVOLTAIC INSTALLATION & MAINTENANCE LESSON NOTES

THIRD TERM

WEEK 1: COMPREHENSIVE REVISION I (THEORY)

1.1 INTRODUCTION

Welcome to your final term. This term is dedicated entirely to consolidating your three years of training and ensuring you are 100% prepared for your final SSCE (WAEC/NECO) examinations. This week is a master-class review of all *theoretical* knowledge from SS1, SS2, and SS3.

1.2 SS1-SS3 THEORY CONSOLIDATION

PILLAR 1: CORE ELECTRICAL & SOLAR PRINCIPLES (SS1)

- **The Photovoltaic (PV) Effect:** The process in a semiconductor (like silicon) where light (photons) strikes the material, knocking electrons loose and creating a flow of Direct Current (DC) electricity.
- **AC vs. DC:**
 - **Direct Current (DC):** Produced by panels, stored by batteries. Electrons flow in one direction.
 - **Alternating Current (AC):** Used by grid/home appliances. Electrons change direction (e.g., 50 times/sec, 50Hz).
- **Ohm's Law:** The fundamental relationship: **Voltage (V) = Current (I) x Resistance (R).**
- **Power Law:** The law of electrical work: **Power (W) = Voltage (V) x Current (I).**
- **Energy vs. Power:**
 - **Power (Watts, W):** The *rate* at which energy is used (e.g., a 100W light bulb).
 - **Energy (Watt-hours, Wh):** The *total amount* of power used over time (e.g., a 100W bulb left on for 5 hours = 500 Wh).
- **Series vs. Parallel Wiring:**
 - **Series (+ to -): Voltage adds up.** Current stays the same. (Used for PV strings and batteries).

- **Parallel (+ to +, - to -): Current adds up.** Voltage stays the same. (Used for PV strings and batteries).

PILLAR 2: COMPONENTS & HARDWARE (SS1-SS3)

- **PV Panels:**

- **Monocrystalline:** High efficiency (18-23%), expensive, black, single crystal.
- **Polycrystalline:** Lower efficiency (15-18%), cheaper, blue, multiple crystals.
- **Bifacial (SS3):** Generates power from *both* sides. Gains power from reflected light (Albedo).

- **Charge Controllers:**

- **Function:** The "brain" for battery systems. Protects from overcharging and deep discharge.
- **PWM (Pulse Width Modulation):** Cheaper, less efficient "switch." Panel voltage must match battery voltage.
- **MPPT (Maximum Power Point Tracking):** More expensive, 30% more efficient "smart converter." Can take high-voltage PV strings and convert down to battery voltage.

- **Batteries (Storage):**

- **Deep-Cycle Lead-Acid (AGM, Gel):** Cheap, heavy, low DoD (~50%), short cycle life (500-1000 cycles).
- **Lithium-ion (LiFePO4) (SS2):** New standard. Expensive, light, high DoD (80-100%), long cycle life (3000-7000 cycles).
- **BMS (Battery Management System) (SS2):** The "brain" for Lithium batteries. *Essential.* Prevents over/under voltage, balances cells, manages temperature.
- **Flow Batteries (SS2):** Industrial. Uses liquid electrolyte. 100% DoD, 20+ year life. Power (stack) and Energy (tank) are separate.

- **Inverters:**

- **Function:** Converts DC power to AC power.
- **PSW (Pure Sine Wave):** Clean, high-quality power. Essential for all modern electronics (TVs, fridges, PCs).

- **MSW (Modified Sine Wave):** "Blocky," dirty power. Cheaper, but can damage sensitive appliances.
- **Mounting & Trackers (SS3):**
 - **Fixed Mount:** Standard roof/ground mount.
 - **Solar Trackers:** Follow the sun. **Single-Axis** (E-W) gives +25-35% yield. **Dual-Axis** (E-W, N-S) gives +45% yield. Expensive and complex.

PILLAR 3: SYSTEM TOPOLOGIES & APPLICATIONS (SS2)

- **Off-Grid (Standalone):** No grid. Requires Panels + Controller + Battery + Inverter. Total independence, high cost.
- **On-Grid (Grid-Tied):** No battery. Requires Panels + Grid-Tied Inverter. Reduces bills. **Does not work during a grid outage** (Anti-Islanding).
- **Hybrid:** "Best of both." Requires Panels + Hybrid Inverter + Battery + Grid. Can run from solar, battery, or grid. Provides backup.
- **Solar Water Pumping:** Uses a **VFD (Pump Controller)**, not a charge controller. Uses a **water tank** as the "battery." Sized by **Total Dynamic Head (TDH)** and **Flow Rate (m³/day)**.
- **Solar Street Lights:** A complete standalone system (Panel, Battery, LED, Controller). Controller uses panel voltage to detect **Dusk-to-Dawn**.
- **Solar Home Systems (SHS):** Small, plug-and-play kits. Often use DC-only loads (lights, phone charging). Enabled by the **PAYG (Pay-As-You-Go)** business model.

PILLAR 4: POLICY & ENVIRONMENT (SS3)

- **Key Agencies:**
 - **REA (Rural Electrification Agency):** Drives rural projects (SHS, Mini-Grids).
 - **NERC (Nigerian Electricity Regulatory Commission):** The *regulator*. Sets rules for **Net-Metering** (selling power back to grid) and Mini-Grids.
 - **SON (Standards Organisation of Nigeria):** Sets product quality standards (SONCAP).
- **Climate Change:**

- **Mitigation:** Reducing CO2 (e.g., 1 solar system replaces 1 generator).
- **Adaptation:** Using solar to adapt to climate effects (e.g., solar pump for drought).
- **Environmental Impact (LCA):**
 - **Manufacturing:** Uses energy and chemicals.
 - **End-of-Life (E-Waste):** The big problem. Panels contain heavy metals (lead). Landfills are dangerous.
 - **Solution: Recycling** (the "Circular Economy").

PILLAR 5: THE BUSINESS OF SOLAR (SS3)

- **Proposals:** The *sales document* given to a client. Must include Load Audit, Solution, BOQ, Price, Warranty.
- **Tenders:** Formal *bids* for government/corporate jobs (Technical Bid + Financial Bid).
- **Costing/BOQ:**
 - **Direct Costs:** All materials (panels, inverter, wires).
 - **Indirect Costs (Overhead):** Your rent, salary, transport.
 - **Pricing:** Your **Total Cost + Profit Margin = Final Price.**
- **Customer Service:** The key is **managing expectations** (e.g., "This system will *not* power your 3 ACs at night"). The **Handover** (training the client) is crucial.
- **Project Management:** Balancing the "Iron Triangle": **Time, Cost, Scope.**

EVALUATION (WEEK 1)

1. Differentiate between a PWM and an MPPT charge controller.
2. Explain the key differences (DoD, Cycle Life, Cost) between Lead-Acid and LiFePO4 batteries.
3. What is "Net-Metering," and which Nigerian agency regulates it?
4. A 5kW pump is installed. What is the "rule of thumb" for the minimum size of the PV array?
5. What is the "3-2-1 Rule" of data backup?

ASSIGNMENT (WEEK 1)

1. Write a one-page "cheat sheet" of all the key formulas and acronyms from SS1-SS3.
 - **Formulas:** Ohm's Law, Power Law, Sizing Formulas (Battery, PV Array), FCR (from SS2).
 - **Acronyms:** AC/DC, PWM/MPPT, BMS, LFP, VFD, TDH, PAYG, REA, NERC, SON, LCA, BOQ, RTO/RPO, IoT.



WEEK 2: COMPREHENSIVE REVISION II (PRACTICAL)

2.1 INTRODUCTION

This week reviews all the *practical skills* and *calculations* required of a technician. This is the "how-to" guide.

2.2 SKILL 1: SAFETY (THE FOUNDATION)

- **PPE (Personal Protective Equipment):**
 - **Fall Protection:** Safety Harness (for roof work).
 - **Electrical:** Insulated Gloves (Class 0, 1000V rated).
 - **Physical:** Steel-toe, rubber-sole boots. Hard hat. Safety glasses.
- **Electrical Hazards:**
 - **DC Arc Flash:** *More dangerous* than AC. A 600V DC arc will not extinguish. *Never* disconnect a live DC string. Use a DC breaker.
 - **LOTO (Lockout/Tagout):** The #1 safety procedure. Shut down all breakers (AC, DC, PV) and "lock" them before starting work.
- **Physical Hazards:** Falls (ladders/roofs), lifting (use legs), heat (sunstroke).

2.3 SKILL 2: SYSTEM SIZING (THE 5-STEP CALCULATION)

This is the *most important* practical "thinking" skill. **SCENARIO:** Size an off-grid system for a client.

- **STEP 1: LOAD ASSESSMENT**
 - List all client's appliances, their Power (W), and Hours/Day.
 - Total Energy (Wh) = Power (W) x Hours (h)
 - Sum them all up. (e.g., Total Load = **5,000 Wh/day**).
 - Find "Peak Load" (all appliances on at once). (e.g., Peak Load = **1,200 W**).
- **STEP 2: INVERTER SIZING**
 - Based on *Peak Load (W)*.
 - Inverter Size = Peak Load x 1.25 (Safety Factor)

- $1,200 \text{ W} \times 1.25 = 1,500 \text{ W}$.
- **Decision:** Choose the next size up (e.g., a **2.5kVA** or **2,000W** Inverter).
- **STEP 3: BATTERY BANK SIZING**
 - Based on *Total Energy (Wh)*.
 - Total Wh to Store = (Daily Energy x Days of Autonomy) / (Battery DoD x System Voltage)
 - Ah Capacity = $(5,000 \text{ Wh} \times 1 \text{ day}) / (0.80 \text{ DoD [Lithium]} \times 24\text{V})$
 - Ah Capacity = $5,000 / 19.2 = 260 \text{ Ah}$.
 - **Decision:** We need a **24V, 260Ah** (or larger) battery bank.
- **STEP 4: PV ARRAY SIZING**
 - Based on *recharging the battery daily*.
 - Array Size (Wp) = Daily Energy / (Peak Sun Hours x System Losses)
 - Array Size (Wp) = $5,000 \text{ Wh} / (5 \text{ PSH} \times 0.80 \text{ efficiency})$
 - Array Size (Wp) = $5,000 / 4 = 1,250 \text{ Wp}$.
 - **Decision:** We need a **1,250 Wp** (or larger) array (e.g., 4 x 350W panels = 1400Wp).
- **STEP 5: CHARGE CONTROLLER SIZING**
 - Based on *PV Array power and voltage*.
 - Controller Amps = Total Array Power / System Voltage
 - Controller Amps = $1400 \text{ Wp} / 24\text{V} = 58.3 \text{ A}$.
 - Add safety factor: $58.3 \text{ A} \times 1.25 = 72.9 \text{ A}$.
 - **Decision:** We need an **MPPT** controller with a rating of **80A** (or 100A).

2.4 SKILL 3: INSTALLATION & WIRING

- **Commissioning (Power-On) Sequence:** This is a critical safety step. **WRONG ORDER DESTROYS EQUIPMENT.**
 1. Connect **Battery** to **Charge Controller**. (Controller "wakes up" and sees 24V).
 2. Connect **PV Array** to **Charge Controller**. (Controller "sees" the high PV voltage and starts charging).
 3. Connect **Battery** to **Inverter**.
 4. Turn on **Inverter** to power AC loads.

- **Advanced Wiring (SS2):**

- **Conduits & Trunking:** Protects wires from sun (UV) and physical damage.
- **Crimping Lugs:** *Essential* for thick battery cables. Use a hydraulic crimper. A bad crimp (using pliers) = high resistance = heat = **FIRE**.
- **Earthing (Grounding):** *Essential* for safety. Panel frames, mounting racks, and inverter chassis *must* be connected to an earth rod.
- **Voltage Drop:** Loss of power in long cables. *Fix:* Use thicker cables, or *increase system voltage* (e.g., 400V string).

2.5 SKILL 4: MAINTENANCE & TROUBLESHOOTING

- **Preventive:**

- **Panel Cleaning:** *Most important.* Use soft brush, water, early morning.
- **Terminal Check:** *Most critical safety check.* Use a **torque wrench** to tighten all terminals (breakers, inverter, batteries) annually. A loose terminal = fire.

- **Corrective (Troubleshooting):**

- **Logic:** Client -> Inverter Screen (Error Code?) -> Breakers -> Battery -> Panels.
- **Battery Test:** Use a Multimeter. Check "Resting Voltage" (e.g., 25.4V). Apply load and see if voltage "crashes" (sign of a dead battery).
- **Panel Test:** Use a Multimeter. Disconnect string (LOTO!). Test **Voc (Open Circuit Voltage)**. If Voc = 0V, you have an open circuit (loose MC4, broken wire).

- **Monitoring (SS3):** Using IoT/apps to see real-time data (PV yield, Battery SoC, Load) and diagnose faults remotely.

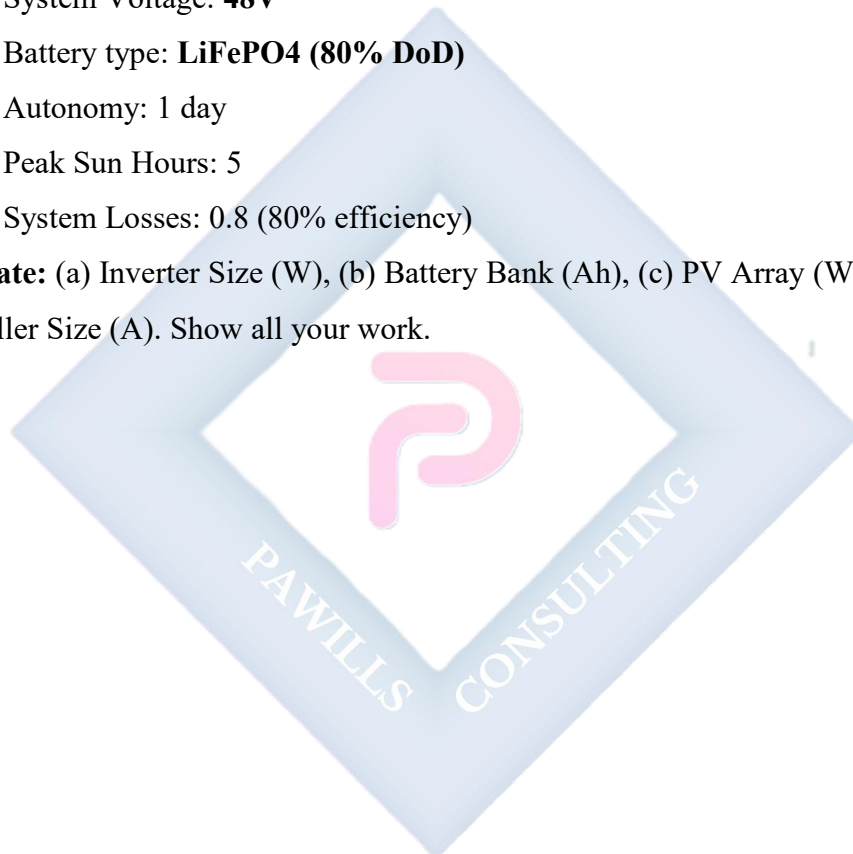
EVALUATION (WEEK 2)

1. What is the **#1 safety rule** for an installer working on a roof?
2. What is the **#1 safety rule** for an installer *before* touching any DC wiring?
3. Why is a "bad crimp" on a battery cable so dangerous?
4. What is the correct, safe "power-on" sequence for an off-grid system?

5. A client's battery is at 24V, but the inverter screen is blank and shows 0V. What is the *most likely* cause? (Hint: Troubleshooting).

ASSIGNMENT (WEEK 2)

1. Using the 5-Step Sizing method, **calculate the specifications** for the following system:
 - Total Daily Load: **8,000 Wh/day**
 - Peak Power Load: **2,000 W**
 - System Voltage: **48V**
 - Battery type: **LiFePO4 (80% DoD)**
 - Autonomy: 1 day
 - Peak Sun Hours: 5
 - System Losses: 0.8 (80% efficiency)
2. **Calculate:** (a) Inverter Size (W), (b) Battery Bank (Ah), (c) PV Array (Wp), (d) Controller Size (A). Show all your work.



WEEK 3: FINAL PROJECT PRESENTATION

3.1 INTRODUCTION

This week, you will prove your competence by presenting a final project. This project combines your technical skills (design, sizing) and your business skills (costing, proposals). You will work in groups or individually.

3.2 PROJECT OPTIONS (CHOOSE ONE)

OPTION 1: THE BUSINESS PROPOSAL (TECHNICAL-BUSINESS)

- **Task:** Create a full, 5-page professional proposal for a *real* (or realistic) client.
- **Client:** A small local business (e.g., a "pure water" factory, a 3-storey school, a 10-room hotel).
- **Deliverable:** A formal proposal document that includes:
 1. **Cover Letter & Introduction.**
 2. **Client Needs Analysis:** A full **Load Assessment** (your calculation).
 3. **Proposed Solution:** (e.g., "A 10kVA Hybrid Solar-Diesel System...").
 4. **System Specifications:** (List of components: Inverter size, Battery kWh, Panel Wp).
 5. **Bill of Quantities (BOQ):** An itemized cost list.
 6. **Performance & Warranty:** (Managing expectations).
 7. **About Your Company:** (Why you are the best).

OPTION 2: THE PRACTICAL BUILD (TECHNICAL-PRACTICAL)

- **Task:** Design and build a small, *functional* Solar Home System (SHS) or a portable power pack.
- **Components:** (e.g., 50W panel, 12V 20Ah battery, small PWM controller, DC lights, USB charging port).
- **Deliverable:**
 1. The *physical, working system* (mounted on a board).

2. A 1-page "User Manual" for your product.
3. A 1-page BOQ showing your costs.
4. A 10-minute demonstration.

OPTION 3: THE RESEARCH REPORT (TECHNICAL-ANALYTICAL)

- **Task:** A 5-page research report on a high-level topic.
- **Topic (Choose one):**
 - "The Business Case for Mini-Grids in Rural Nigeria: A Case Study of [Your LGA]."
 - "The E-Waste Crisis: A 10-Year Plan for Solar Panel Recycling in Nigeria."
 - "A Comparative Analysis: Why Lithium (LiFePO₄) is now cheaper than Lead-Acid for solar."
- **Deliverable:** A formal report with research, data, and conclusions.

3.3 PRESENTATION

- Each student/group will have 10-15 minutes to present their project to the class and instructor (acting as the "client" or "examiner").
- You must be professional, clear, and able to *defend* your calculations and design choices.

3.4 ASSESSMENT CRITERIA

You will be graded on:

1. **Technical Accuracy:** Are the calculations correct? Is the design sound?
2. **Professionalism:** Is the proposal/report well-written and neatly presented?
3. **Completeness:** Did you include all the required elements (e.g., BOQ, wiring diagram, warranty)?
4. **Verbal Presentation:** Were you clear, confident, and able to answer questions?

EVALUATION / ASSIGNMENT (WEEK 3)

- Your assignment is to **complete and present your chosen Final Project.**

WEEK 4: MOCK EXAMS

4.1 INTRODUCTION

This week is dedicated to a full-scale Mock Examination to simulate the SSCE experience. This will test your knowledge, time management, and "exam-fu" (your ability to write what the examiner wants to see).

4.2 EXAM STRATEGY

- **Objectives:** Read the question, then the options. Eliminate the two "stupid" answers first. Choose the *best* answer from the remaining two. Answer every question.
- **Theory/Essay:**
 - **Verbs:** "List" = bullet points. "Explain" = paragraphs. "Differentiate" = table.
 - **P.E.E. Method:** **Point** (Make your statement), **Explain** (Describe *why*), **Example** (Give a real-world example).

4.3 MOCK EXAMINATION (SAMPLE PAPER)

SECTION A: OBJECTIVE (Answer all 10)

1. What is the main advantage of an MPPT charge controller over a PWM? (A) It is cheaper. (B) It is lighter. (C) It can convert high PV voltage to low battery voltage. (D) It works only with lead-acid batteries.
2. A solar street light turns on at night by sensing: (A) A drop in panel voltage. (B) A drop in battery voltage. (C) A signal from a motion sensor. (D) A timer.
3. The *most important* safety equipment for working on a roof is: (A) A hard hat. (B) Insulated gloves. (C) A safety harness. (D) Steel-toe boots.
4. A client needs 4,000 Wh of energy per day. They are using a 24V system with 50% DoD lead-acid batteries. What is the *minimum* battery bank size required? (A) 166 Ah (B) 333 Ah (C) 208 Ah (D) 416 Ah

5. Which of these is *not* a function of a Battery Management System (BMS)? (A) Cell balancing (B) Over-voltage protection (C) Increasing the battery's voltage (D) Over-temperature protection
6. "RPO" (Recovery Point Objective) is a measure of: (A) How fast the system can be repaired. (B) How much *time* the system can be down. (C) How much *data* the business can afford to lose. (D) The total profit of the project.
7. A client's inverter screen is blank, and the system is dead. The *first* tool you should use, and the *first* component you should test is: (A) A clamp meter on the PV array. (B) A multimeter on the battery terminals. (C) A laptop to check the inverter software. (D) A torque wrench on the AC breaker.
8. In a "Solar-as-a-Service" business model for a factory, the factory (client) pays for: (A) The full cost of the installation upfront. (B) The *energy (kWh)* they use, at a rate cheaper than diesel. (C) The cost of the panels only. (D) A monthly rental fee for the inverter.
9. What is the *most common* cause of a "No Service" fault on a GSM phone? (A) A faulty PMIC. (B) A corrupt IMEI. (C) A bad SIM card. (D) A damaged or disconnected antenna cable.
10. A PC's BIOS reports "1 long, 3 short" beeps. This is a standard code for: (A) A RAM failure. (B) A CPU failure. (C) A Video Card failure. (D) A Power Supply failure.

SECTION B: THEORY/ESSAY (Answer 3 of 4)

Q1 (Safety & Maintenance): (a) What is "LOTO" (Lockout/Tagout), and why is it a critical safety procedure? (b) Explain the difference between "Preventive Maintenance" and "Corrective Maintenance." (c) List and describe three (3) essential Preventive Maintenance tasks.

Q2 (Sizing & Troubleshooting): (a) What is "Total Dynamic Head (TDH)" in a solar water pumping system? (b) A client's solar street lights are all failing after 18 months. What is the *most likely* component to have failed, and *why*? (c) A client's inverter is beeping with an "Overload" alarm. What is the likely cause?

Q3 (Business & Policy): (a) Differentiate between the role of **NERC** and the role of **REA** in Nigeria. (b) You are preparing a proposal for a client. List the three (3) main components of the

"Cost" calculation (e.g., Total Price = A + B + C). (c) Explain the "PAYG" (Pay-As-You-Go) business model.

Q4 (PC/GSM Repair): (a) What is "rooting" an Android phone, and what is the single biggest *risk* of doing it? (b) A client's phone shows "Fake Charging." What component is the most likely culprit? (c) A client's PC is slow. Explain the *three main hardware upgrades* you could recommend, in order of effectiveness.

(Mock Answers: 1C, 2A, 3C, 4B (8000/24=333), 5C, 6C, 7B, 8B, 9D, 10C)



WEEK 5: MIDTERM TEST (MOCK EXAM ANALYSIS)

5.1 ANALYSIS OF MOCK EXAM

(This week is designated as "Midterm Test" in the scheme, but logically it should be the analysis of the Mock Exam from Week 4.)

The "Lesson" for this week is to review your marked Mock Exam script.

1. **Identify Your Weakest Areas:** Go through your wrong answers. Where did you lose marks?
 - **Calculations?** (e.g., the sizing question) -> *You must memorize the formulas.*
 - **Definitions?** (e.g., "What is LOTO?") -> *You must use flashcards and learn the keywords.*
 - **Concepts?** (e.g., "Why MPPT > PWM?") -> *You must re-read the notes for that topic.*
 - **Time?** (Did you fail to finish?) -> *You must practice writing faster.*
2. **Focus on Question "Verbs":**
 - Did you *list* when it asked you to *explain*? (You lost easy marks).
 - Did you *explain* when it asked you to *list*? (You lost precious time).
 - Did you *differentiate* (use a table) when asked?

5.2 TARGETED REVISION

Based on your mock results, you will now create a **Personalized Study Plan**.

- **Activity:** List your 5 "weakest" topics. (e.g., "1. Sizing formula, 2. NERC vs REA, 3. GSM Software, 4. PC Beep Codes, 5. Project Management").
- **Assignment:** For this week, your *only* assignment is to re-write your *worst* essay answer from the mock exam, using your notes, and make it a "perfect" 10/10 answer.

WEEK 6: FINAL SSCE PREPARATION

6.1 THE "TOP 20" FINAL CRAM SHEET

This is the final countdown. Below are the Top 20 concepts you must know.

1. **Safety (PPE/LOTO):** #1 priority. Harness for roof, gloves for electricity. LOTO to de-energize.
2. **AC vs. DC:** AC (Grid/Appliances), DC (Panels/Batteries).
3. **Ohm's/Power Law:** $V=IR$, $P=IV$.
4. **Series vs. Parallel:** Series (V adds), Parallel (I adds).
5. **Panel Types:** Mono (efficient, black) vs. Poly (cheaper, blue).
6. **Controller Types:** PWM (cheap) vs. MPPT (smart, efficient).
7. **Battery Types:** Lead-Acid (50% DoD, 2-yr life) vs. LiFePO4 (80-100% DoD, 10-yr life).
8. **BMS:** *Essential* for Lithium. Protects and balances.
9. **Inverter Types:** PSW (Pure, for all) vs. MSW (Modified, for simple loads).
10. **System Topologies:** Off-Grid, On-Grid (no backup, anti-islanding), Hybrid (best of all).
11. **The 5 Sizing Steps:** Load $\rightarrow$ Inverter $\rightarrow$ Battery $\rightarrow$ Array $\rightarrow$ Controller.
12. **The 4 Commissioning Steps:** Battery $\rightarrow$ Controller $\rightarrow$ PV $\rightarrow$ Inverter.
13. **Key App: Pumping:** VFD Controller, Water Tank = Battery. Sized by TDH + Flow.
14. **Key App: Street Light:** Dusk-to-Dawn (senses panel voltage). Failure = Battery.
15. **Key App: SHS:** Plug-and-play, DC loads, PAYG model.
16. **Key Regulators:** REA (rural projects), NERC (rules, net-metering), SON (product quality).
17. **Key Maintenance:** Clean panels, **Torque terminals** (prevent fire).
18. **Key Troubleshooting:** Use DMM. Check Battery V, then check Panel Voc.
19. **Key Business:** Cost (Direct+Indirect) + Profit = Price. Manage expectations.
20. **PC/GSM:** "No Boot" (BIOS, RAM/GPU beeps). "No Data" (APN). "No Network" (IMEI, Antenna cable).

6.2 FINAL EXAM ADVICE

- **Get Sleep:** A tired brain cannot solve a calculation problem.

- **Bring Tools:** Have your calculator (non-programmable), pens, and pencils.
- **Show Your Work:** For any calculation (e.g., sizing), write the **Formula**, the **Substitution**, and the **Answer**. You get marks for the formula even if the math is wrong.
- **Be Confident:** You have completed three years of training. You have the knowledge.
Good luck.



WEEK 7: MIDTERM BREAK

This is the final break before the external examinations.



WEEKS 8–10: WAEC/NECO PRACTICAL & THEORY EXAMS

This period is not for lessons. This is the official window for the Senior Secondary Certificate Examinations. Your practical and theory exams will be held during this time.



WEEKS 11–13: CLOSING

- **Final Review:** (Post-exam) Discussion of the SSCE questions.
- **Career Talk:** Discussing "What's Next?"
 - University (Electrical Eng., Renewable Energy)
 - Polytechnic (Electrical/Electronics)
 - **Apprenticeship:** The *fastest* way to a career. Get a job as an "apprentice" with a professional installation company.
 - **Entrepreneurship:** Starting your *own* business (start small!).
- **Closing:** Final collection of records and graduation.

